Reverse engineering and hacking Ecovacs robots

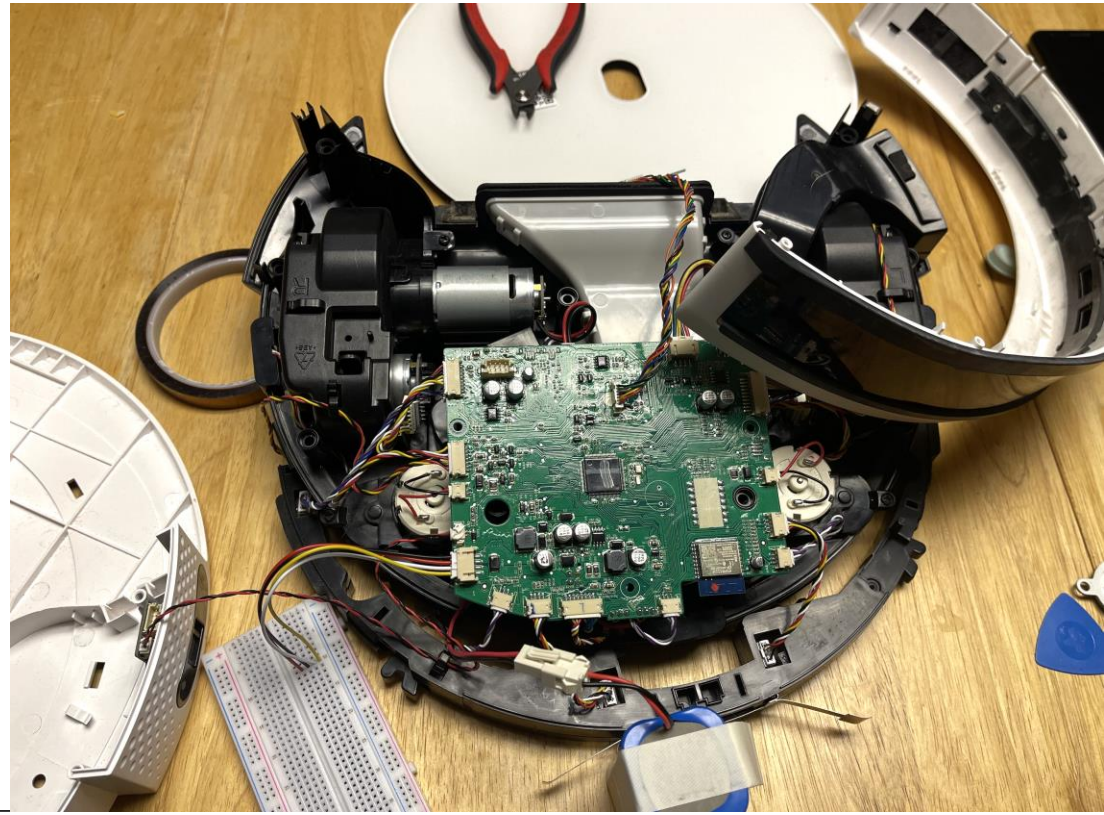
DEFCON 32 – Dennis Giese, braelynn

About Dennis

- “Security Researcher” aka Hardware Hacker
  - Research field: Wireless and embedded Security&Privacy
- Interests: Reverse engineering of interesting devices
- Vacuum Robot (and IoT) collector
  - Rooting of vacuum robots
  - https://robotinfo.dev

About braelynn

- Hacks things for Leviathan Security Group
  - (this talk is entirely personal research and does not reflect their views ;))
- Focus: Application Security and APIs
- Started robot hacking during COVID
- Now: Hardware hacking for fun
  - Robots, Cameras, Smart locks



Goals of this talk

- Understand Security & Privacy risks of IoT devices
- Get an overview of vacuum robot hacking
- Learn about vulnerabilities and how to find them
- Ultimate goal: get root access without disassembly

Past research on robots

- CCC Congress 34C3 (2017)
- DEFCON 26 (2018)
- DEFCON 29 (2021)
- DEFCON 31 (2023) + CCC Camp 2023
- CCC Congress 37C3 (2023)

Disclaimers

- We do not claim that any vendors use sensors to spy on you!
  - (but they can in theory)
- We cover primarily physical attacks (or proximity attacks) on devices
- Many vendors have problems
  - Independent of origin, size, market share
  - This talk: Ecovacs
- Research part of private projects
  - No sponsorship by companies or organizations
  - Any statements are our own opinion and not representing any organization

Devices covered in this Talk

- Ecovacs DEEBOT 900 Series
- Ecovacs DEEBOT N8/T8 *
- Ecovacs DEEBOT N9/T9 *
- Ecovacs DEEBOT N10/T10
- *Ecovacs DEEBOT X1 **
- Ecovacs DEEBOT T20 *
- *Ecovacs DEEBOT X2 **
- Ecovacs Goat G1
- *Ecovacs ~~Spybot~~ Airbot Z1*
- Ecovacs Airbot AVA
- Ecovacs Airbot ANDY
- Yeedi *

We will only focus on devices that run Linux.

Device with Cameras
Device with Microphone / "Hi YIKO"
* = wildcard

About this talk

- Result of 5 years of research and experiments
- Not the first/only ones researching Ecovacs robots
- The vendor knows about most of our findings
  - … they tried to fix some and failed
  - … seem to ignore the worst vulnerabilities

Shoutout to Chris Anderson (@0xHexHijinx)

Collaborative effort

@tihmstar and Dennis hacking Robot cameras at NULLCON Goa 2023

Lawnmower hacking at CCC Camp 2023 in the ZTL/N.O.R.T.x village with Dennis, Maurice, Axel L., Micha B., Mona, Antre (no picture, because forgot to make one :/)

https://nullcon.net/
https://events.ccc.de/camp/2023/hub/camp23/en/assembly/nort/

MOTIVATION

Why do we want to root devices?

- Play with cool hardware
- Stop devices from constantly phoning home
- Use custom Smart Home Software
- Diagnosis of broken devices
- Verification of privacy claims

Why do we not trust IoT?

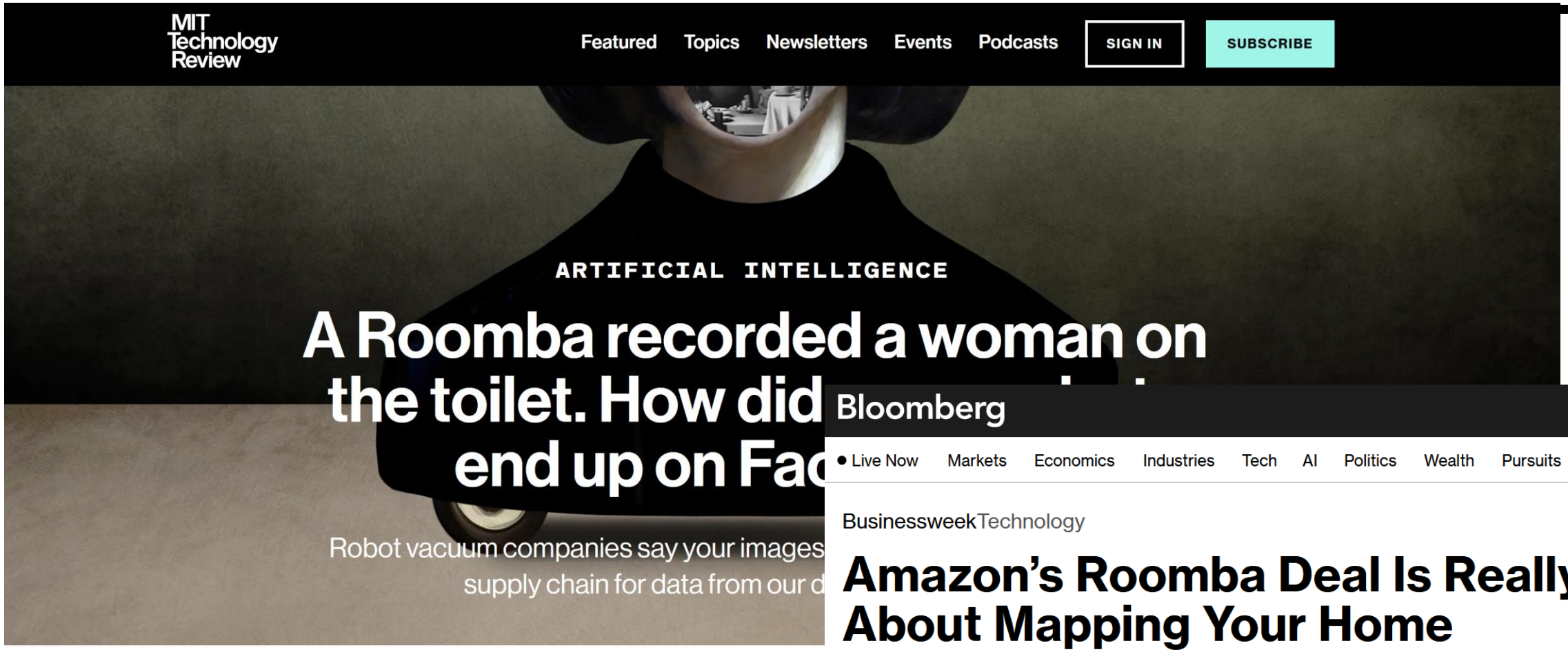
- Devices are connected to the home network
- Communication to the cloud is encrypted, content unclear
- Developing secure hardware and software is hard
- Vendors get caught with shady behavior

Microphones??

Cameras

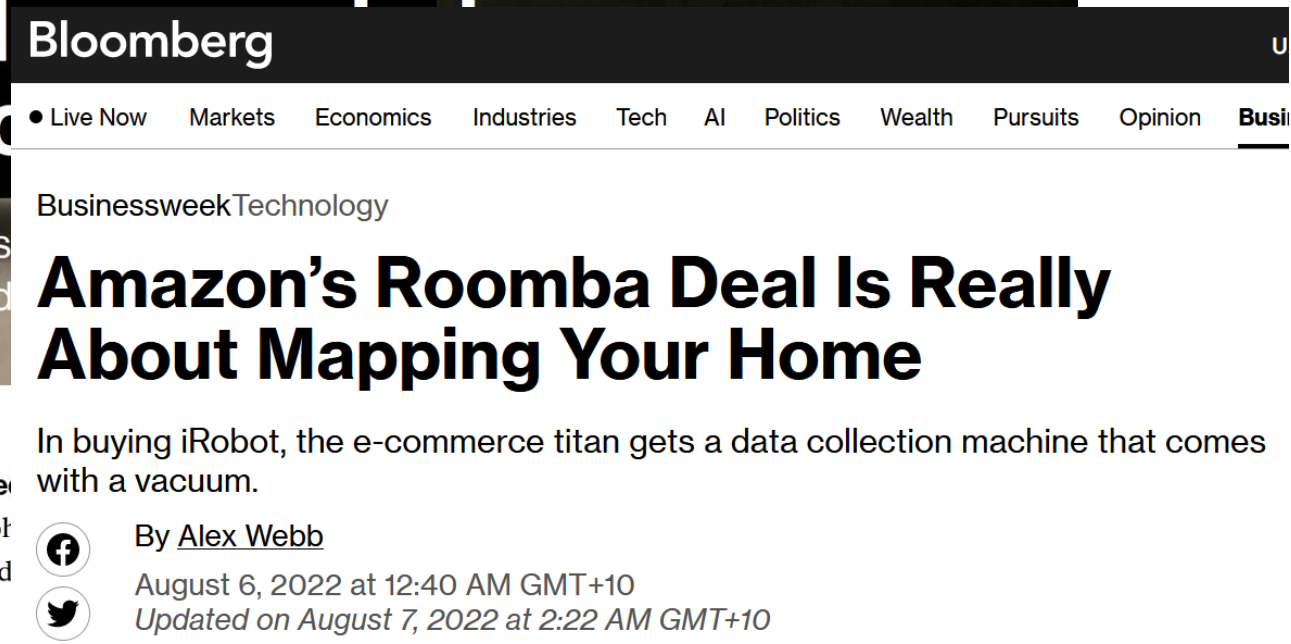
Risks of devices with cameras

- Devices might store pictures indefinitely … and some do. both cloud and local
- Used devices might be problematic
  - Previous owner installed rootkit
  - New owner cannot verify software
  - Result: Device might behave maliciously on your network
- Root access is the only way to verify that a device is „clean“



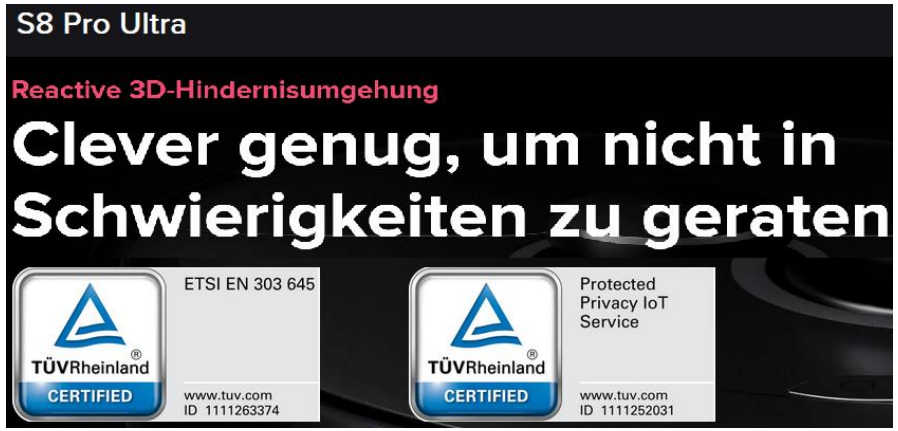
by **Eileen Guo**
December 19, 2022

In the fall of 2020, gig workers in Venezuela poste
forums where they gathered to talk shop. The ph
sometimes intimate, household scenes captured

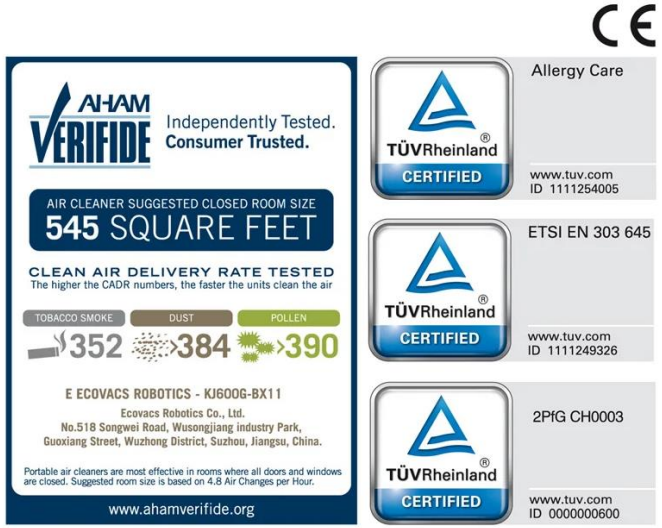


https://www.technologyreview.com/2022/12/19/1065306/roomba-irobot-robot-vacuums-artificial-intelligence-training-data-privacy/
https://www.bloomberg.com/news/articles/2022-08-05/amazon-s-irobot-deal-is-about-roomba-s-data-collection#xj4y7vzkg

Can you rely on Certifications?



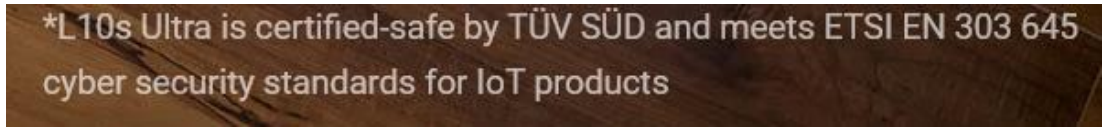
Source: https://de.roborock.com/pages/roborock-s8-pro-ultra



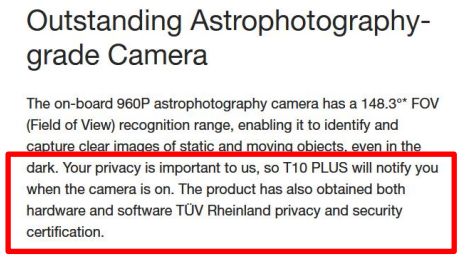
Source: https://www.ecovacs.com/global/airbot-air-purifier-robot/airbot-z1

Source: https://www.mi.com/global/product/xiaomi-robot-vacuum-x10-plus/

Source: Ecovacs iOS application loading screen



Source: https://www.dreametech.com/products/dreamebot-l10s-ultra

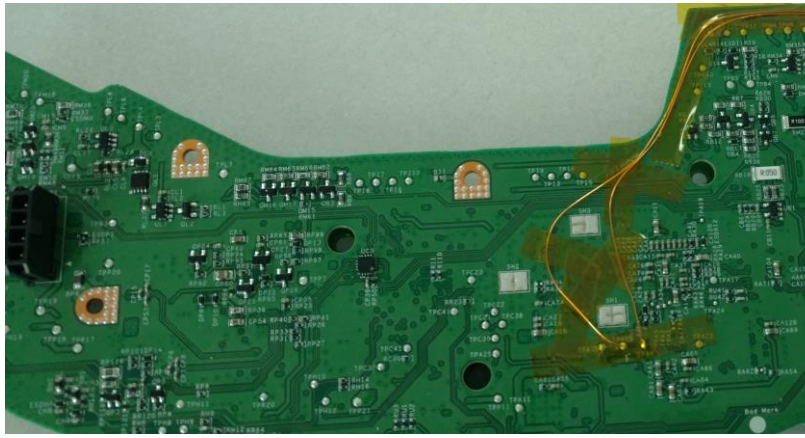


Source: https://www.ecovacs.com/global/deebot-robotic-vacuum-cleaner/deebot-t10-plus

ROBOT HACKING JOURNEY

First work in 2017

- 34C3 (2017) and DEF CON 26 (2018)
  - Work together with Daniel Wegemer
  - Targets: Xiaomi Vacuum Robot / Roborock S5
  - OTA broken, local updates possible
- Released in 2019
  - Targets: Roborock S6, S5 Max, S7 and others
  - Custom bootloader tool, requiring teardown



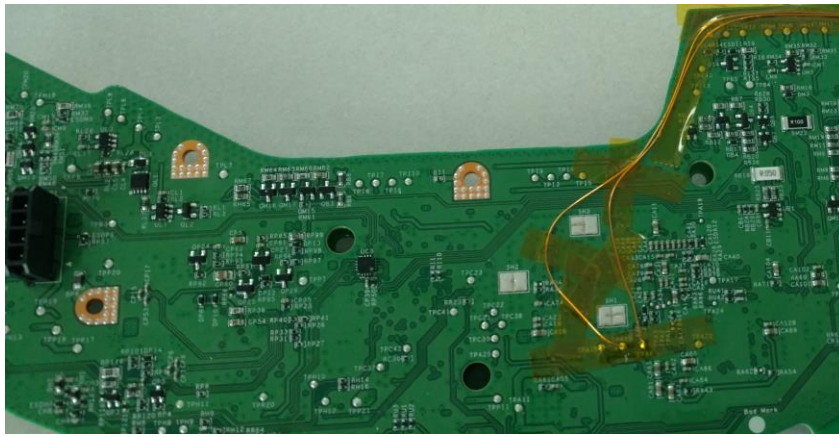
First look at Ecovacs (2018)

- After CCC talk in 2017:
  - Ecovacs Deebot 900 from an influencer
  - early firmware with debug symbols
- Findings:
  - Firmware unprotected, TLS broken, no integrity protection
  - Device can be rooted by MITM by using malicious OTA
  - Problem: hardware extremely weak
- Results were never published as project was abandoned

Really bad youtube video of teardown:
https://www.youtube.com/watch?v=BrM9JJoX72g

New rooting methods

- DEFCON29 (2021)
  - Targets: Roborock S6 MaxV, Dreame L10
  - Bypass Secure boot and security features
- DEFCON31 and CCC Camp 2023
  - Targets: Dreame, Roborock, Narwal, Shark
  - Secure Boot bypass thru bootloader magic
- Issue: finding new rooting methods becomes annoying
  - Solution: let's attack a new vendor



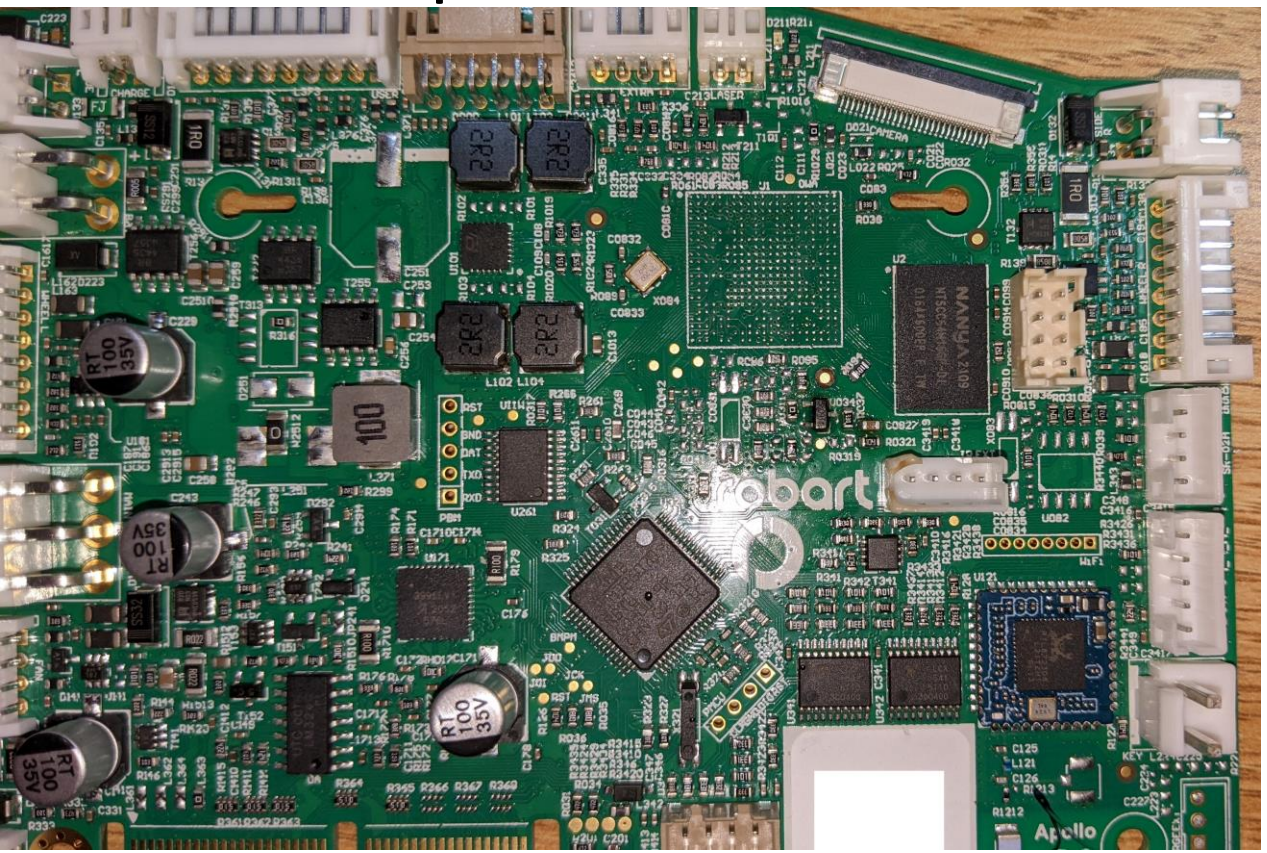
https://dontvacuum.me/talks/DEFCON29/DEFCON29-Robots_with_lasers_and_cameras.html
https://dontvacuum.me/talks/DEFCON31/

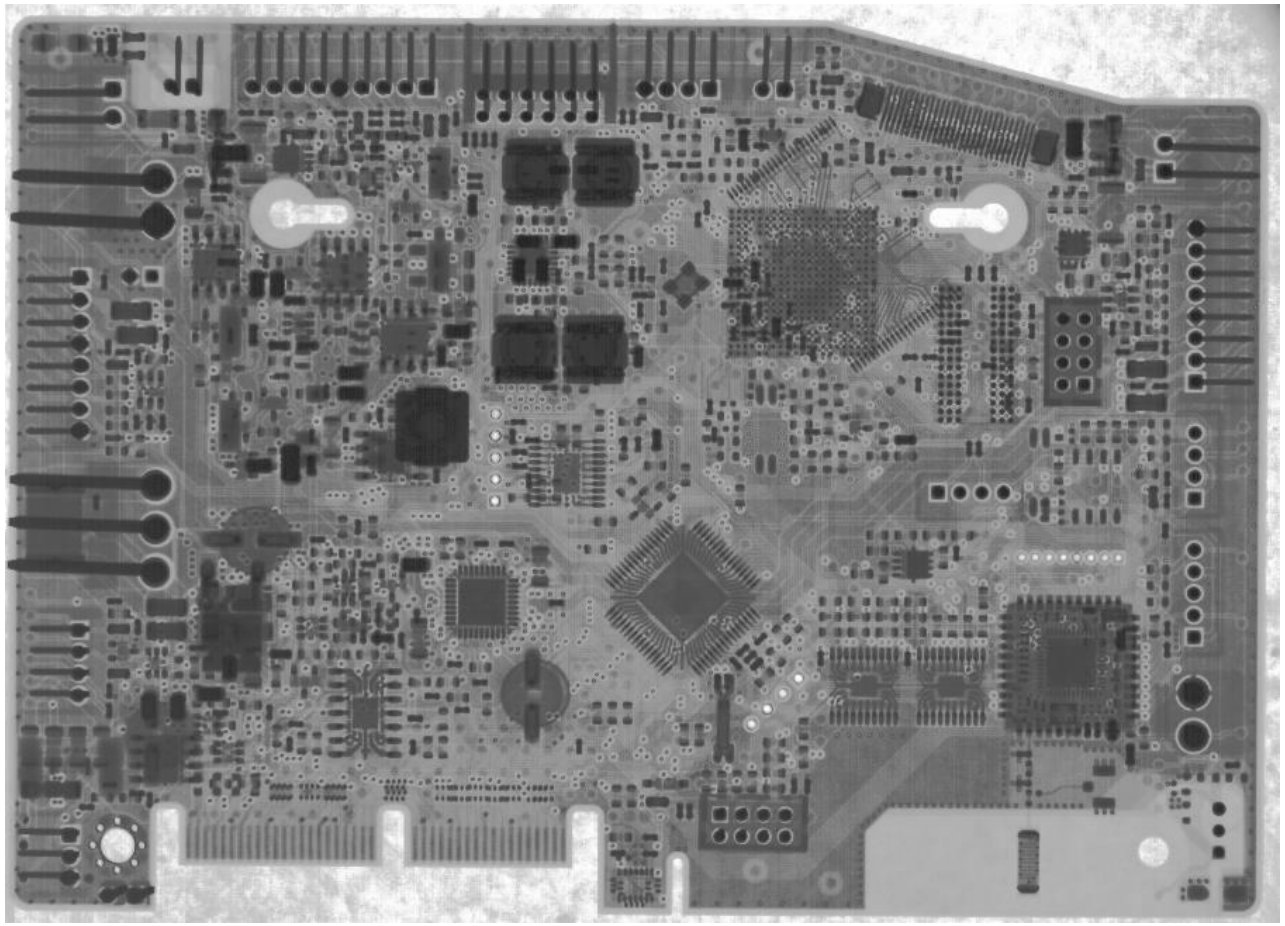
Taking a look again on Ecovacs (2021)

- Ecovacs releases more powerful models
- Features very similar to competition, but with lower price
- Analyzed device: Deebot X1
  - Time to root: 30 minutes
  - Archivement: modified filesystem
- Potential target for more rooting efforts in the future
- Independently: braelynn hacked Ecovacs and Yeedi robots

De-obfuscation of obfuscation With X-Rays

- Example: Shark Robot





ECOVACS ECOSYSTEM

Why Ecovacs?

- Founded in 1998 in Suzhou, China
  - Original intent: production of OEM vacuum cleaners
- Introduction of their flagship model “Deebot” vacuum in 2007
- 17% market share in 2020, second to iRobot
  - Global market share is likely higher now
  - Currently, Ecovacs market cap is 5x higher than iRobot’s

Early rendition of Deebot

Image source: https://www.ecovacs.com/us/newsroom/official-news/ecovacs-25th-anniversary
Data source: Statista

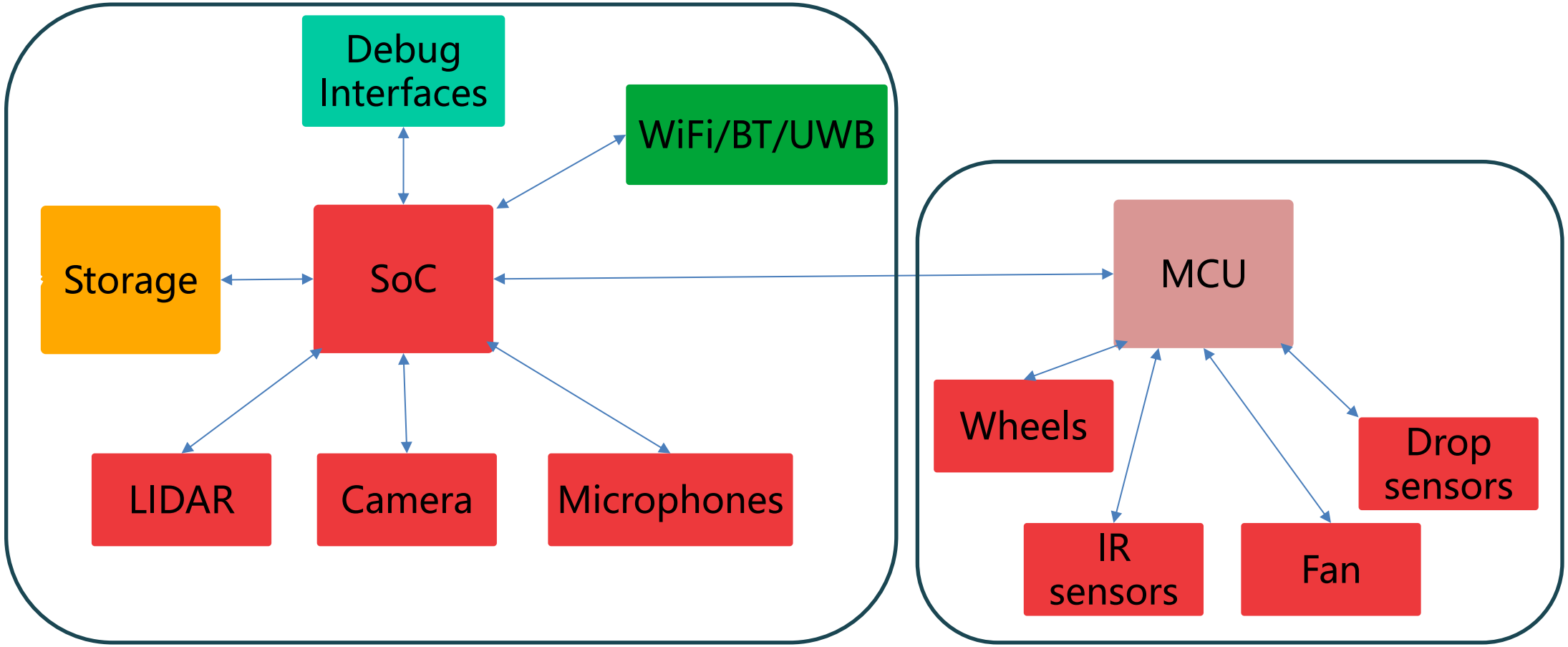
Products

More than 130 SKUs!

Source: Ecovacs.com and Ecovacs Home App

HARDWARE

Hardware



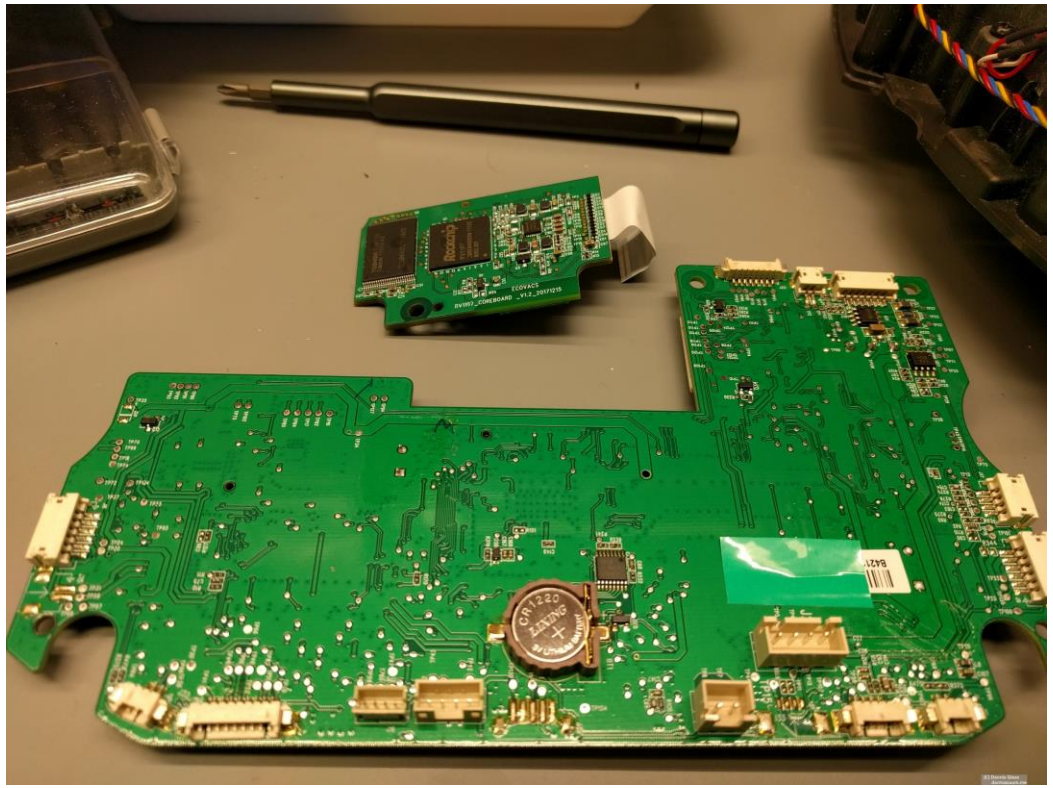
Linux/Navigation/Mapping/Connectivity

Realtime operations

*Not all device models have Cameras/Lidars/Microphones

Hardware: Deebot 900 series

- Released 2018
- Based on Rockchip RV1107
  - 1 ARM cores, 128 Mbyte RAM
- 256 MByte NAND Flash
- Sensors:
  - LIDAR
  - IR sensors



Weak hardware: Not interesting for hacking

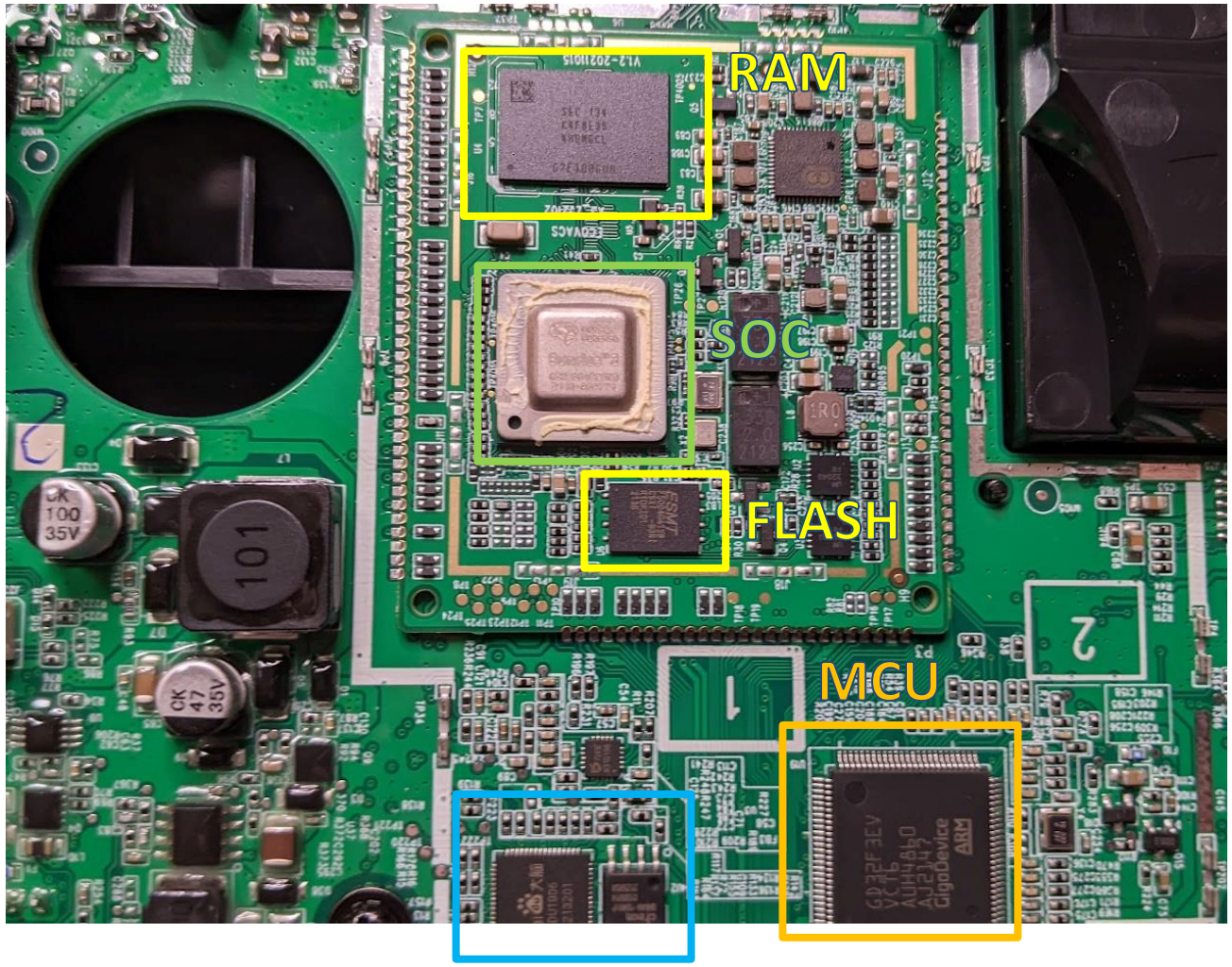
https://dontvacuum.me/teardowns/ecovacs-deebot-900
https://rockchip.fr/RV1107%20datasheet%20V1.1.pdf

Deebot Vacuum robots

- Interesting and cheap hardware
- Similar hardware for multiple generations
- Example X1
  - Features:
    - Station control
    - Voice assistant
    - Remote view

Hardware: Deebot X1

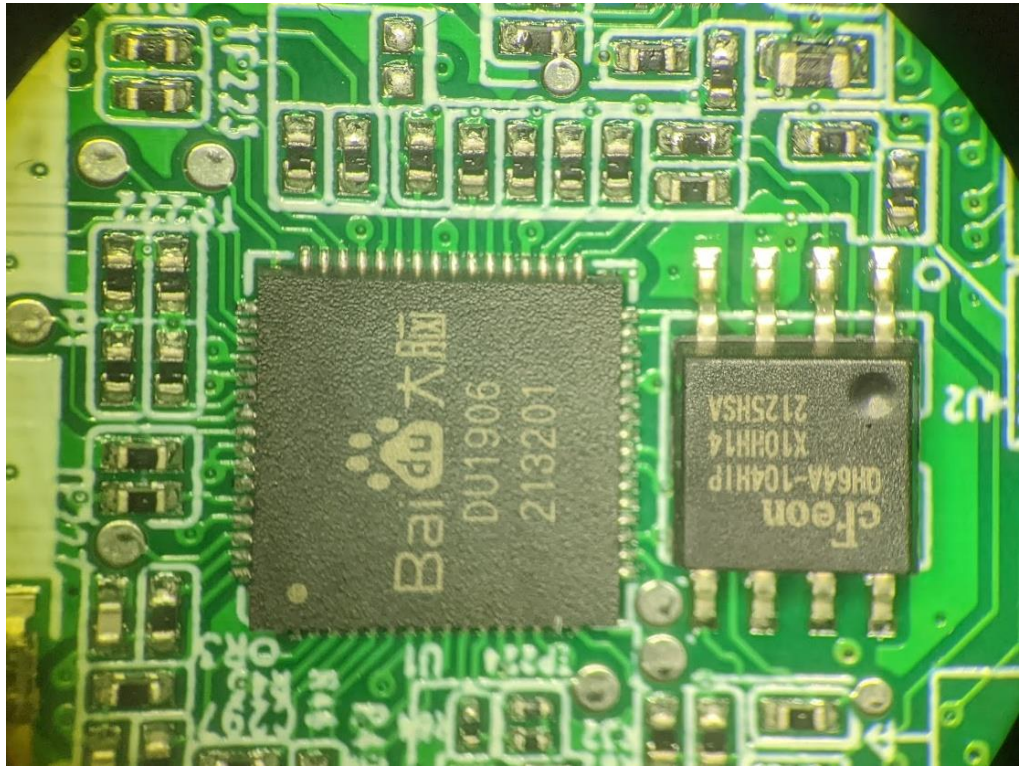
- Same hardware as in X2*
- Based on: Horizon X3 SoC
  - 4x Cortex-A53 processor
  - 1x Cortex-R5 core
  - AI accelerator
- 2 GByte DDR4 RAM
- 512 Mbyte SPI NAND flash
- GD32 MCU



*X2 uses eMMC flash

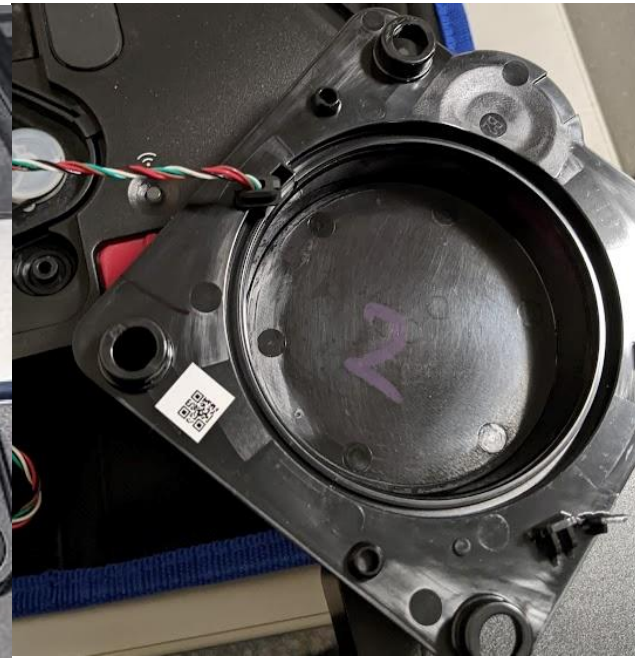
Hardware: Deebot X1

- Special chip: Baidu AI/DSP IC
  - DU1906 voice processing chip
  - Own firmware on SPI flash
  - Wake-up word detection



Hardware: Deebot X1

- Sensors
  - Lidar
  - Microphone array
  - Camera+Line Lasers
  - Lots of IR distance sensors



Airbot Z1

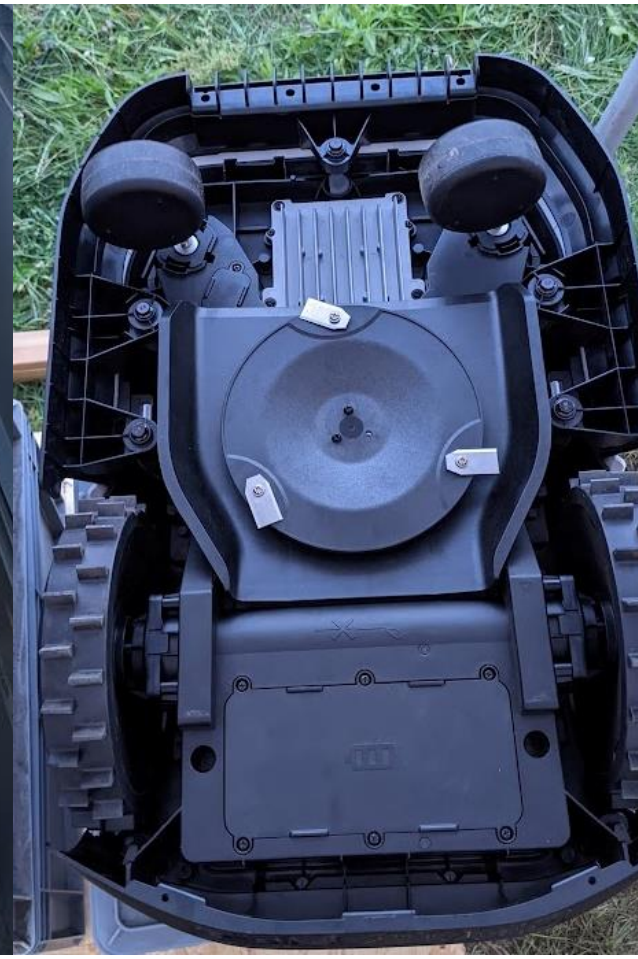
Fun fact: 2-in-1 robot
Deebot X1 + Deebot 900 serie
Connected via Ethernet

- Released 2023
- Based on hardware platform of X1
  - Difference: additional camera
  - 6 Microphones
- Features:
  - Bluetooth speaker
  - Air filter and Humidifier
  - Home Patrol

Source: https://www.ecovacs.com/de/airbot-air-purifier-robot/airbot-z1

Goat G1 Lawnmowing Robot

- Released
  - 2023 in EU, AU
  - 2024 in US (G1-GX)
- Navigation
  - GPS
  - Visual, ToF
  - UWB Beacons
- Features:
  - Optional LTE
  - Remote view/Patrol



Reliable Gardening Security

Binocular cameras capture real-time courtyard images to provide all-around protection.
Your data stays safe and secure with TüV Rheinland-certified data protection.

https://www.ecovacs.com/au/campaign/goat-robotic-lawn-mower

ECOVACS

Hardware: Goat G1

- Based on: Rockchip RK3588
  - 4x Cortex-A76
  - 4x Cortex-A55
  - AI accelerator
- 4 GByte DDR4 RAM
- 16 Gbyte eMMC flash
- Multiple GD32 MCUs
  - Display, Knife assembly

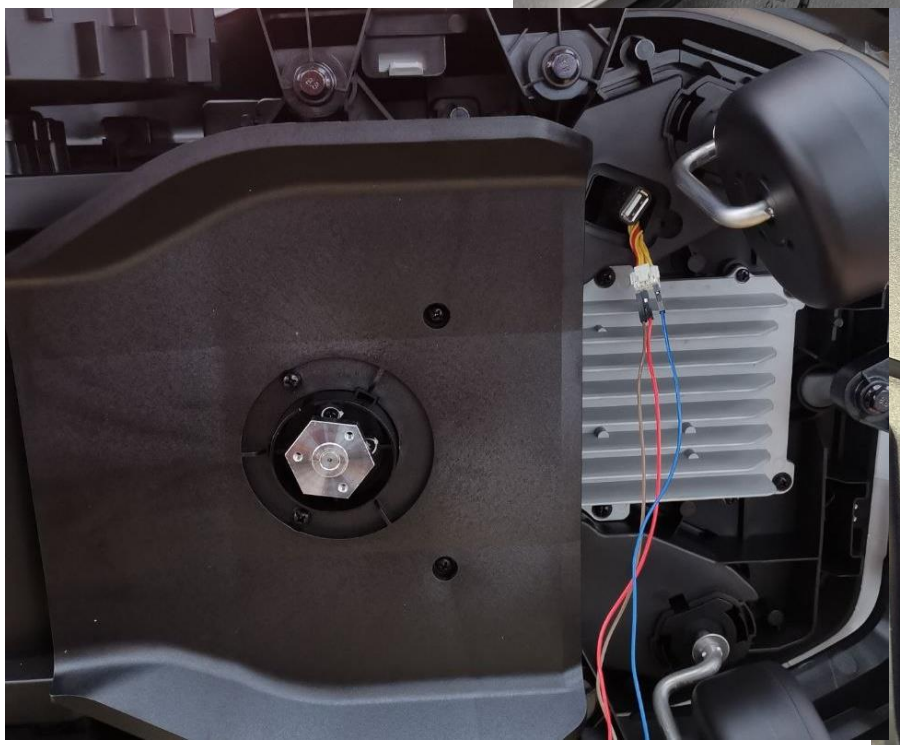
Hardware: Goat G1

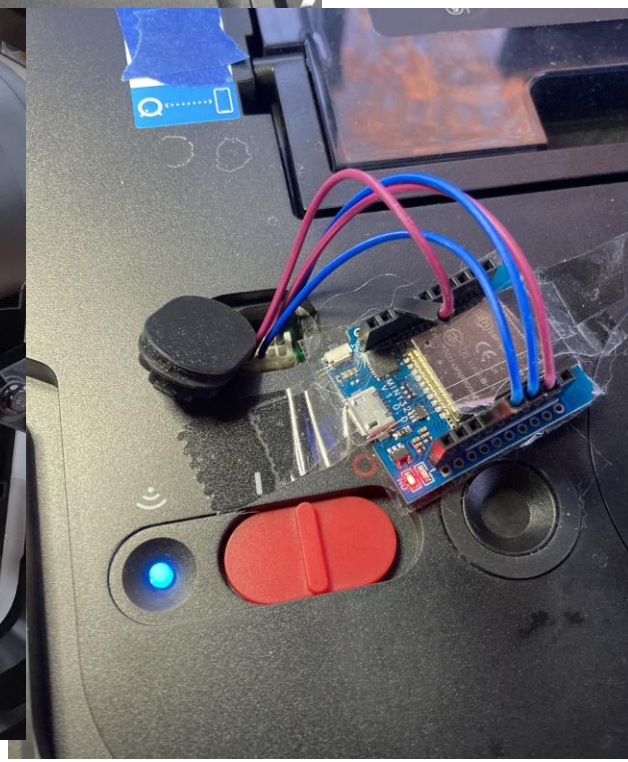
- Sensors
  - 360° Camera (3MP)
  - Front Camera (2MP)
  - ToF Camera
  - Rain detector
  - Bump switches

Debug Ports

- Similar for all models since 2019
- Provides:
  - UART
  - 3.3V
  - SWD
  - USB

Easy to debug and root without breaking warranty seals!





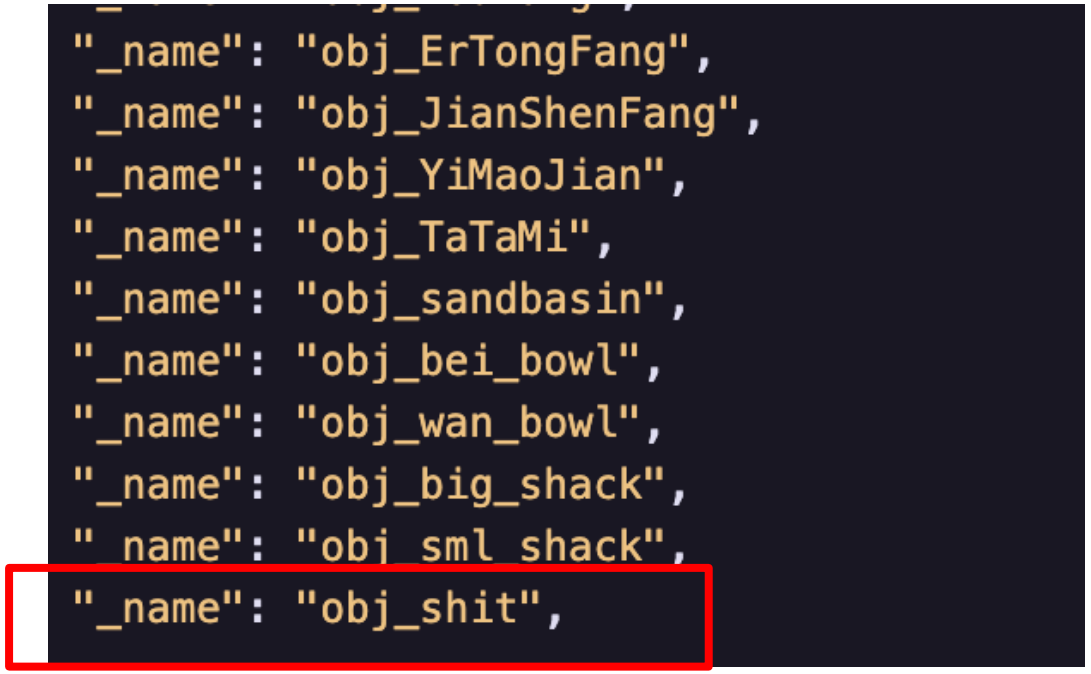
SOFTWARE

Ecovacs Software

- Linux OS
- ROS Melodic Morenia
- Custom Ecovacs software: „Medusa“
- Software packages:
  - Full Python 2.7 environment
  - AWS Kinesis SDK (remote camera access outside of China)
  - Alibaba Aliyun SDK (remote camera access inside of China)
- Good: little or no protections against rooting

AI models

- Tensorflow and OpenCV is used for detection
- Typical objects:
  - Furniture
  - Cable
  - Pets and Pet „remains“
- Lawn Mower:
  - Small animals
  - Face Recon



Firmware updates

Ecovacs Deebot X2 Omni (US)

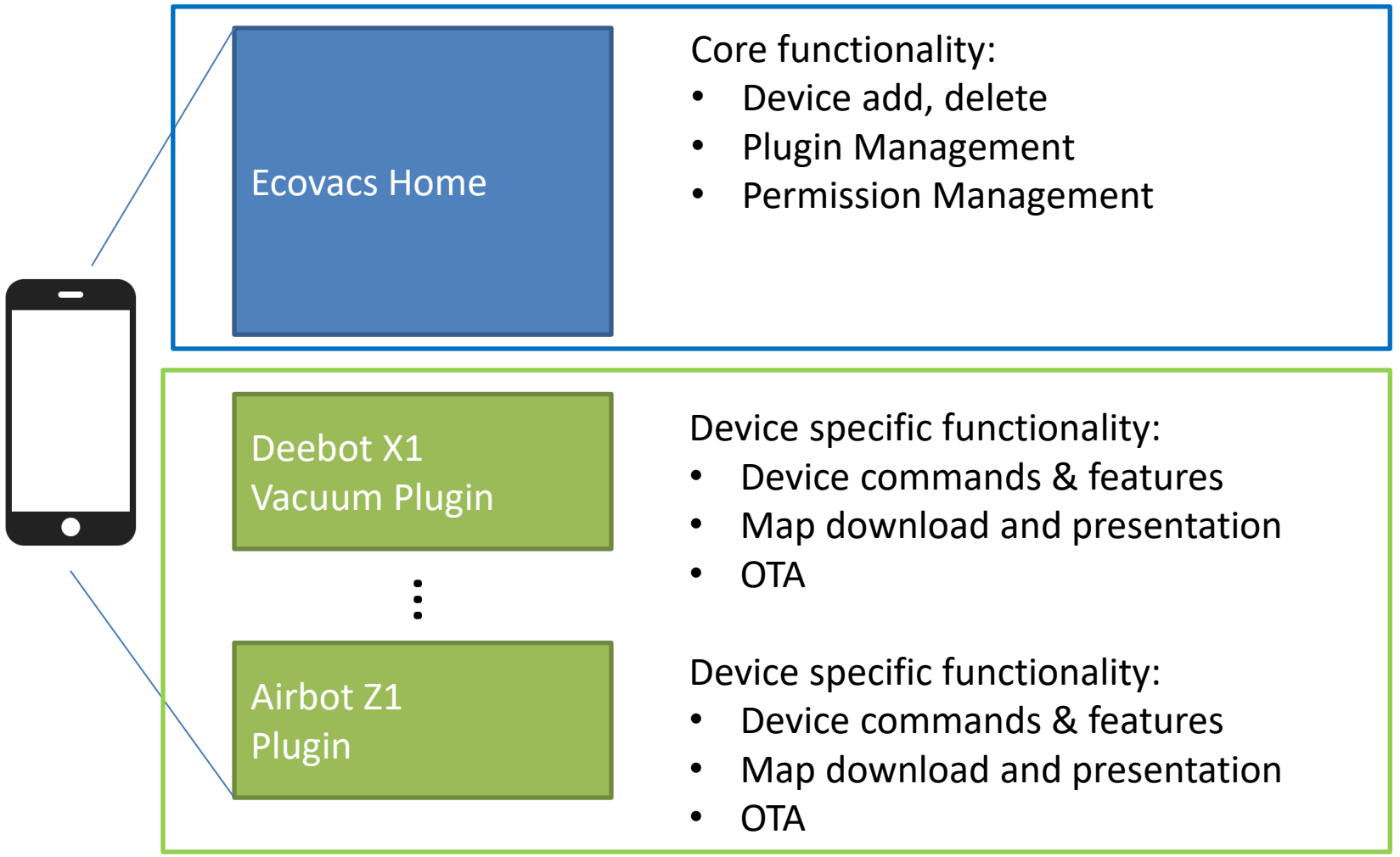
MD5	Filename	Version	Datetime	Size
0036e1d9f9ebb2851849f648429180e7	zj2228_fw-1.39.2.bin	1.39.2	21/08/2023 01:42	236,02 MB
a460c4a940ef5ecd164dc6db75705a60	304c1c01012a33751edd4c4b2bcb987c	1.62.0	20/11/2023 02:34	217,986 MB
aeb41d49fb9d2296a17af64c579e2e61	1e39fa53974426404f6bea2fc343f4a8	1.70.0	29/12/2023 01:20	218,327 MB
bad05c5a5ca96881cbccf4654cf2e0bb	fdeda7cf4bd1ff216677d37dd4b45f02	1.75.0	30/01/2024 02:00	225,764 MB
326dbf455c7442a1c2e3802485e252a8	87014780532514635a5b4aa63710ed95	1.76.0	22/03/2024 02:26	225,764 MB

Oct 2023 Release

Mar 2024 EOL?

Version	Changelog
1.39.2	Fixed known issues and optimized user experience.
1.62.0	Fixed known issues and optimized user experience.
1.70.0	Fixed known issues and optimized user experience.
1.75.0	Fixed known issues and optimized user experience.
1.76.0	Fixed known issues and optimized user experience.

App Structure



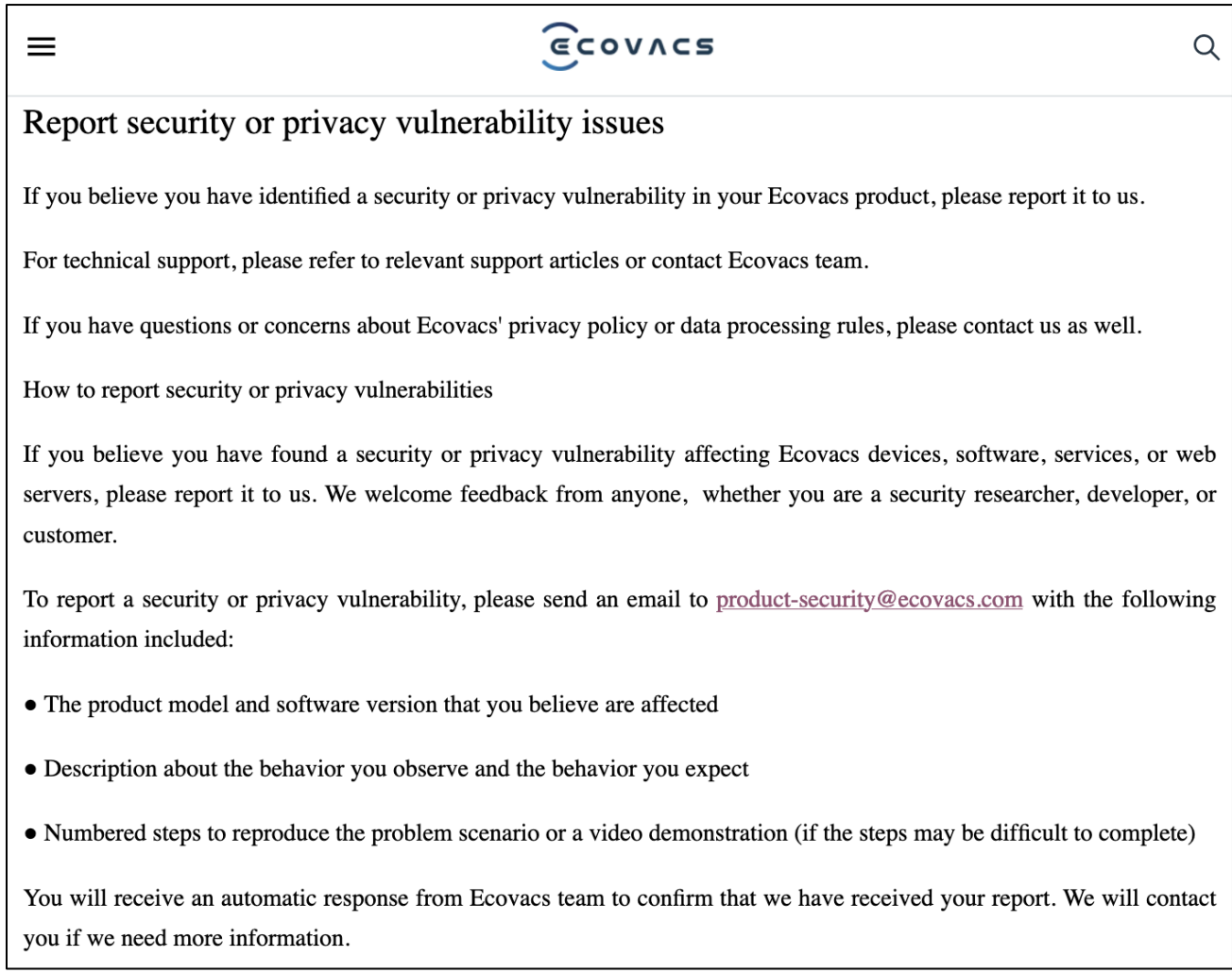
Native Android/iOS app

VueJS code
(downloaded dynamically)

SECURITY & PRIVACY

Ecovacs security

- No real, collaborative bug bounty program
  - promises acknowledgement on public bulletin board
  - …bulletin board doesn't exist?
- Never replied to our reports
  - Same experience for other researchers
- Some bugs are silently patched



ECOVACS

Report security or privacy vulnerability issues

If you believe you have identified a security or privacy vulnerability in your Ecovacs product, please report it to us.

For technical support, please refer to relevant support articles or contact Ecovacs team.

If you have questions or concerns about Ecovacs' privacy policy or data processing rules, please contact us as well.

How to report security or privacy vulnerabilities

If you believe you have found a security or privacy vulnerability affecting Ecovacs devices, software, services, or web servers, please report it to us. We welcome feedback from anyone, whether you are a security researcher, developer, or customer.

To report a security or privacy vulnerability, please send an email to product-security@ecovacs.com with the following information included:

- The product model and software version that you believe are affected
- Description about the behavior you observe and the behavior you expect
- Numbered steps to reproduce the problem scenario or a video demonstration (if the steps may be difficult to complete)

You will receive an automatic response from Ecovacs team to confirm that we have received your report. We will contact you if we need more information.

Source: https://www.ecovacs.com/global/report-issues

Privacy policy

- No guarantees that data stays in user locale
- Generally, regional AWS services used
- Photos and videos sent to Ali Cloud Video for AI analysis in certain models
- Lots of telemetry data collected

Privacy concerns

- Vacuums equipped with microphones and cameras
  - Can they be enabled remotely without user notice?
  - Where is the data sent?
- AI
  - Why do robots need face recon AI models?
  - Is telemetry data being used to train AI?

Privacy concerns

Unauthorized Access to Video Feeds

The first common fear about robot vacuum camera privacy is outsiders gaining unauthorized access to the device's video feed or recordings. In a story that went viral in 2022, pictures of a female sitting on a toilet, captured by a robotic vacuum cleaner, circulated around the Internet. The manufacturer responded by saying that the image had been taken as part of the device's training, but the fact that the image had been captured and made public left a bad taste in people's mouths.

DEEBOT robot vacuums counter hackers accessing cameras by **encrypting all data gathered by the device** (including videos) with the AES-128 (128-bit Advanced Encryption Standard.

Can Robot Vacuum Cameras Be Hacked?

2023-08-14

CONTENTS

1.Why Do Smart Vacuums Have Cameras?
2.What Kind of Data Do Robot Vacuums Gather?

Source: https://www.ecovacs.com/us/blog/robot-vacuum-privacy-concerns

Ecovacs certifications

- Ecovacs boasts multiple security certifications from TÜV Rheinland
  - Claims to meet ETSI 303 645
  - Hardware and software certifications
- Mobile application loading screens advertise ISO/IEC 27001:2013

Source: Ecovacs iOS application loading screen

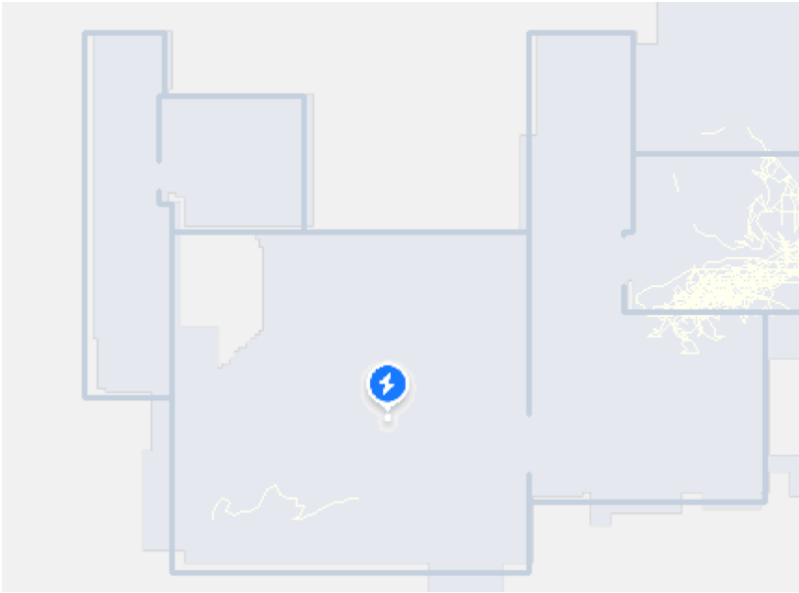
FINDINGS

Data harvesting

- Mobile apps and robots are chatty, a lot of communication with Ecovacs servers
- Key data collection API endpoints: “bigdata”, “data_upload”
- Telemetry data collected
  - Live coordinates of robot location in home
  - Wi-Fi access points, network data
  - Additional information if robot gets stuck
  - AI pictures? (even if not opted in)

Data retention in the Cloud

- Maps and pictures are stored in a NoSQL database
  - Anyone who knows the ObjectID can access the data
- Maps associated with robots seem to remain on servers
  - Survive factory reset
  - Re-pairing to different account
  - Deletion of account has no effect
- Tokens remain valid after account deletion
  - Access to robot still possible



User Data storage on device

- User data partition not encrypted
- Lots of log, configs, maps and pictures stored on partition
  - Live Video pin (MD5 hash), mower pin (plaintext)
  - Wi-Fi credentials, Neighbor Wi-Fi access points
  - “Hello Yiko” traffic logs
- Factory reset: does not fully erase all information
  - Sensitive log files remain
  - Additional problem: flash wear leveling

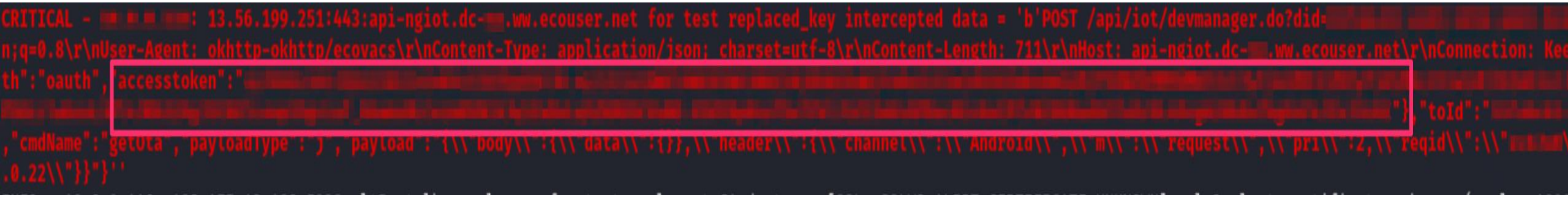
Selling a used device, even if it is factory reset: Risk to your privacy!

Learn about IoT and wear leveling here: https://dontvacuum.me/papers/ACMWisec-2021/

TLS sadness in the App

- Ecovacs Home app correctly checks certificates
  - However, the robot-specific plugins don't always do
- Plugins accept self-signed certificates
  - Risk in insecure Wi-Fi networks, e.g. Airport,Hotel,DEFCON
  - No warning or error shown in App
- Leaks user account auth tokens
  - Allows the attacker full access to account and devices
  - Tokens expire after 7 days

TLS sadness in the App



Encoded PASTE A TOKEN HERE

eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9.ey

Decoded EDIT THE PAYLOAD AND SECRET

HEADER: ALGORITHM & TOKEN TYPE

```
{
  "alg": "RS256",
  "typ": "JWT"
}
```

PAYLOAD: DATA

```
{
  "c": "
  "u": "
  "r": "
  "t": "a",
  "iat": 170339
  "exp": 170399
}
```

TLS sadness in the Robot

- MQTT & TLS connections accept self-signed certs on some devices
  - Allows MITM
  - OTA updates can be injected
  - Perfect tool: certmitm by Aapo Oksman

down_audio_hook.sh

```
#!/bin/sh
…
wget --no-check-certificate -T 60 -O /tmp/${LANGUAGE_ID}.tar.gz_ ${LANGUAGE_URL} && mv /tmp/${LANGUAGE_ID}.tar.gz_ /tmp/${LANGUAGE_ID}.tar.gz
…
```


https://github.com/aapooksman/certmitm

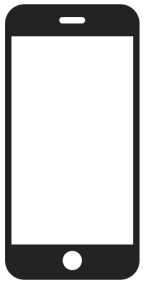
Sneaky live video

- Audio warning when camera is accessed
  - At start of access and every 5 minutes
  - Implementation: sound file is played
- Problem:
  - Localized sound files stored on /data
  - sound files can be deleted or replaced
- Attack: replace warning with empty file

Live video ap(p)ocalypse

- App allows live audio+video access on robot
  - Functionality provided by AWS Kinesis
- Owner and shared users can access feature
- Protected by PIN
  - Asks for PIN before connecting
  - Can only be changed and reset by owner
  - Reset requires account credentials

Live video ap(p)ocalypse



Ecovacs Cloud

PIN 1234

Ask user for PIN

"Verify hash(eco_1234)"

Check if Pin hash matches with local hash

"Pin is correct"

Was the Pin reported as correct?

"Start AWS Kinesis WebRTC session"

Kinesis SDK: Join session

WebRTC session data

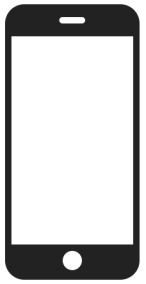
Join WebRTC session

Via AWS Kinesis

Stream video and audio

Ecovacs cloud acts as a relay/broker

Live video ap(p)ocalypse



Ecovacs Cloud

PIN 1234

Ask user for PIN

"Verify hash(eco_1235)"

Check if Pin hash matches with local hash

"Pin is incorrect"

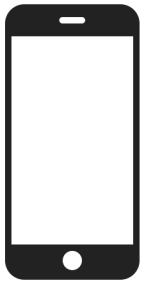
Was the Pin reported as correct?

Show error „Wrong pin!"

Are PIN verification and WebRTC tied together?

Copyright (c) 20th Century Fox Film Corp

Live video ap(p)ocalypse



Ecovacs Cloud

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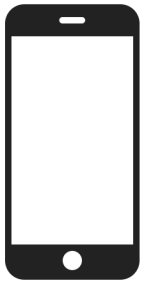


Ecovacs Cloud

PIN 1234

Not caring about any PIN

"Start AWS Kinesis WebRTC session"

WebRTC session data

Kinesis SDK: Join session

Join WebRTC session

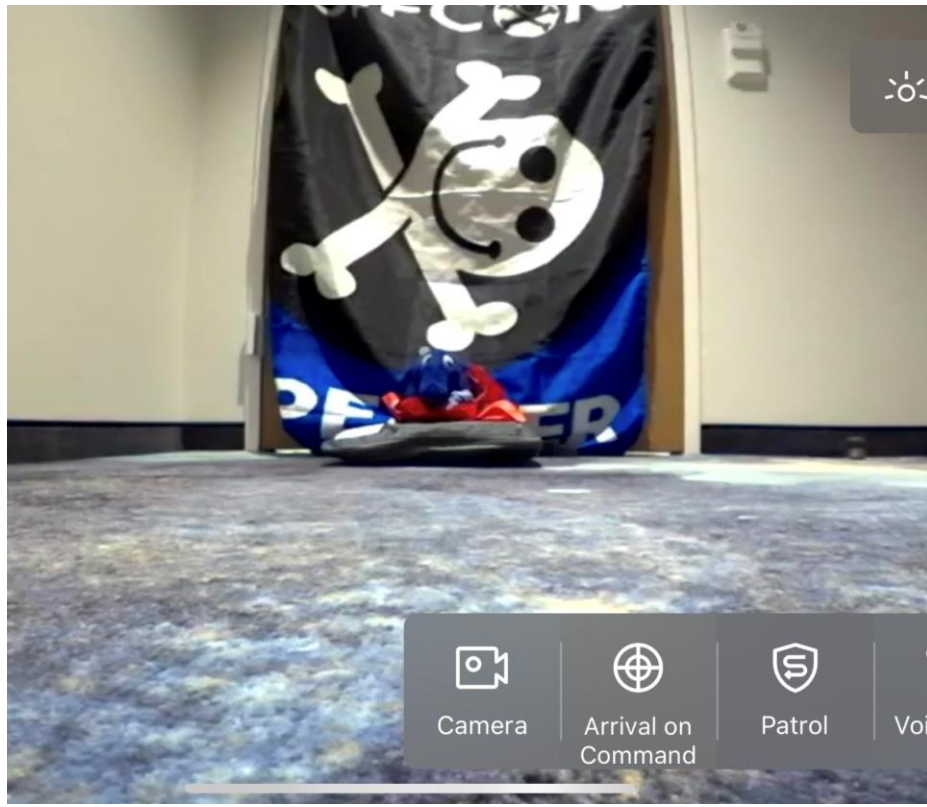
Via AWS Kinesis

Stream video and audio

Note: This is NOT a vulnerability in AWS Kinesis. The issue is in Ecovacs implementation!

Live video ap(p)ocalypse

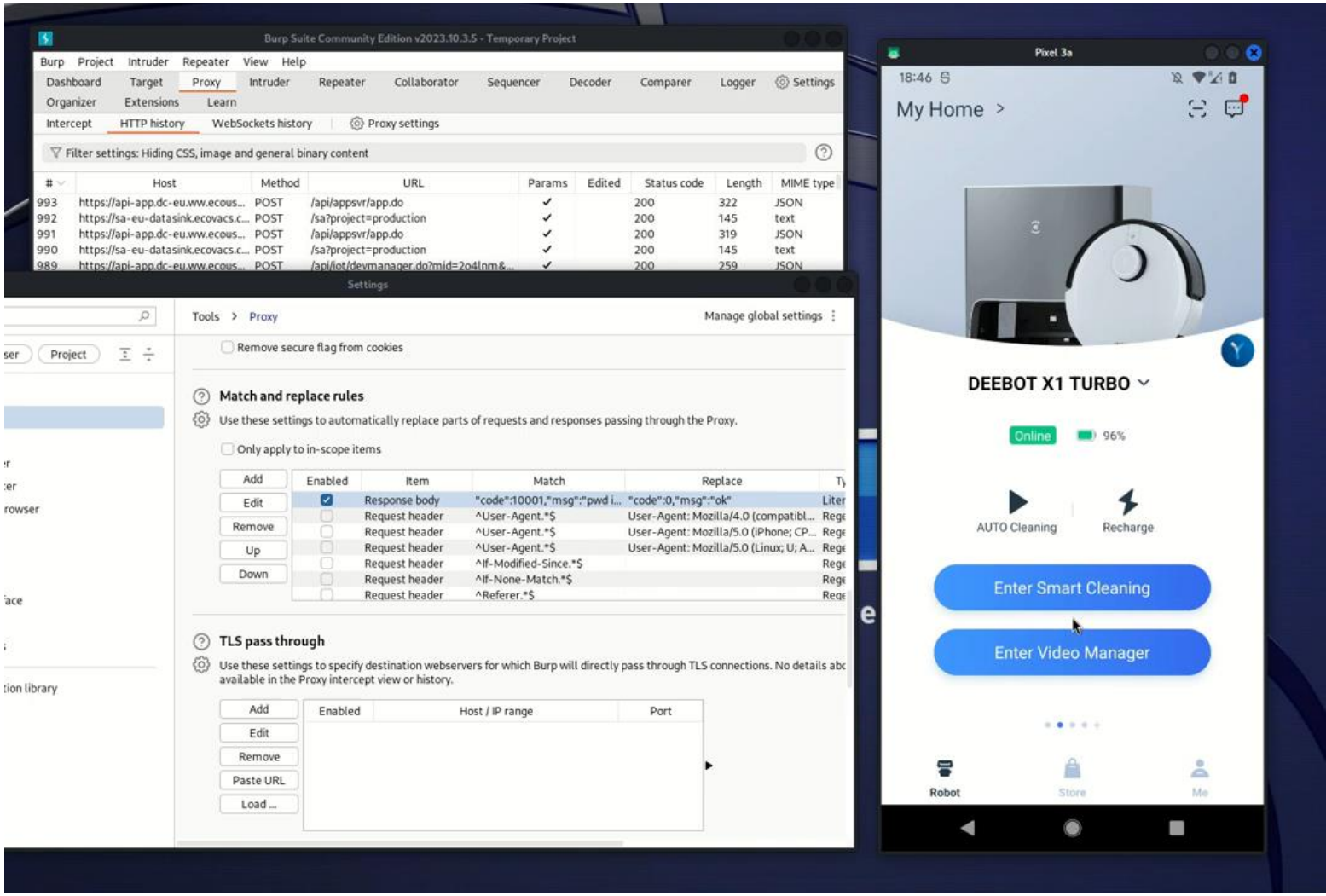
- PIN protection implemented in app
  - Client-based authentication and ACL enforcement
  - Robot does not keep track of successful authentications
- Log of video stream access relies on honesty of app
- Really bad in combination with TLS issue or shared accounts
- Even worse: If sound files have been tampered



„Honor“ system also applies to other aspects in the App

Live video ap(p)ocalypse DEMO

- Reported in 2023
- Unsuccessful fix pushed
  - some plugins updated
  - no firmware fixes
- Downgraded app still works



*Free lawn mowers

- Ecovacs Goat G1 has anti-theft mechanism integrated
  - If robot gets picked up, PIN is required to unlock
  - Alarm can get triggered
- Problem:
  - Protection is implemented in SoC (and not in MCU)
  - Pin is stored in plaintext on robot
- Other mowers have the same issue

Do not keep your mower outside unprotected!

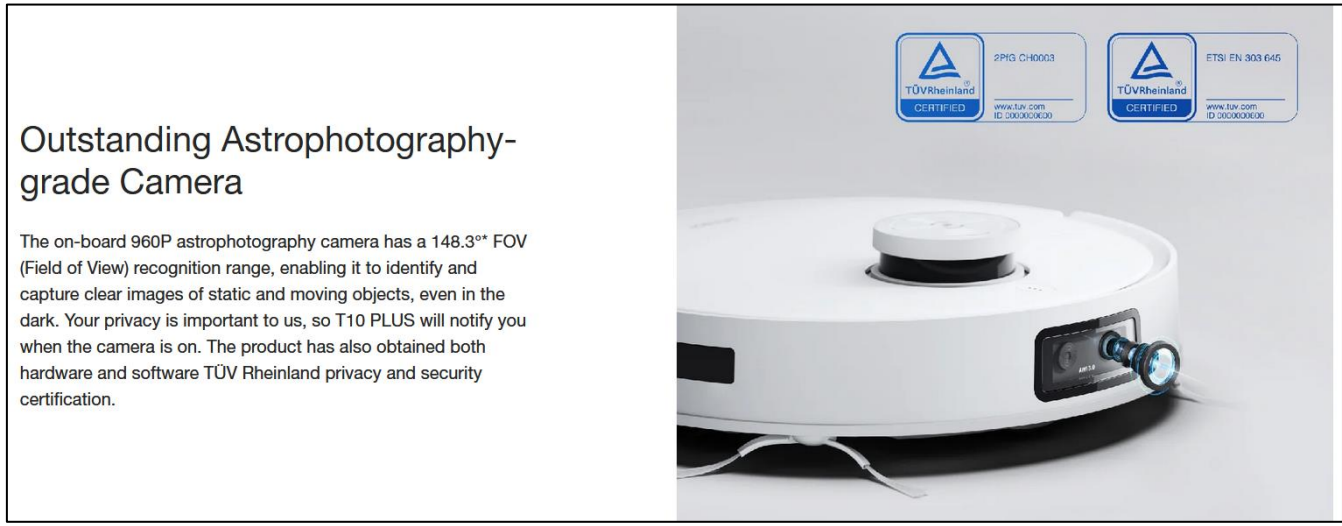
Solution: improvement of locking mechanism

*lawn mower not actually free. Do not steal robots!!!

Bluetooth remote code execution

- Newer vacuum robots and all lawn mowing robots use BLE for provisioning
  - Lawnmowers: BT is always active
  - Vacuums: BT is active for 20 minute after booting / daily reboot
- Communication between app and robots via GATT protocol
  - Payload encrypted with static AES key such as “12345678ecovacs”
  - Input validation is… insufficient

These devices have cameras and microphones



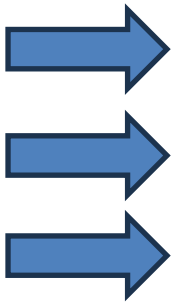
BLE RCE payload creation

- Command for execution: **“/etc/rc.d/play_boot_music.sh start”**
- Embedding in JSON payload
- Generation of encrypted payload
  - **6046506C6F6C2D636F6E7461637420757321313030303030303031004361320071004B6944006ABD4D5CBF917DA0EDEF04AD99A53F9B0D43E5CF6CC1F06DF5AAB803F1BFBD4C5E338DCE48C9B6EBA5C5076020518A4A9367F5F8244D7DF78B91EB6C6BCEBC940F2AEDFBC0000F0A**

BLE RCE payload execution

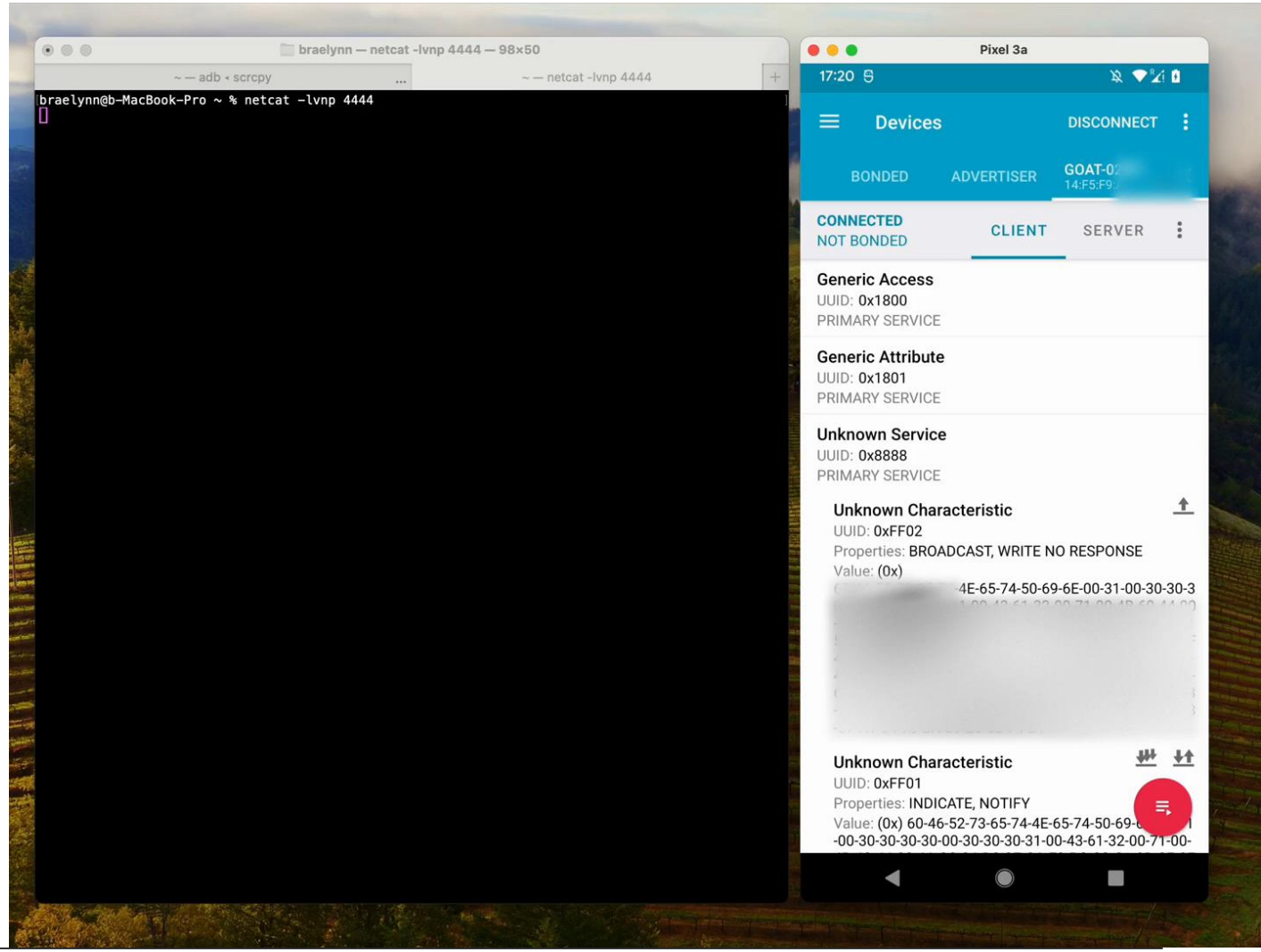
- Robot software receives payload and decrypts it
- Parsing of payload and extraction of “foo”
- Generation of command line, transmit “foo” via environment variable
  - **‘foo="";/etc/rc.d/play_boot_music.sh start; "" /usr/bin/somestuff’**
- Execution of command line

We tested this with 50 meters distance (165ft).



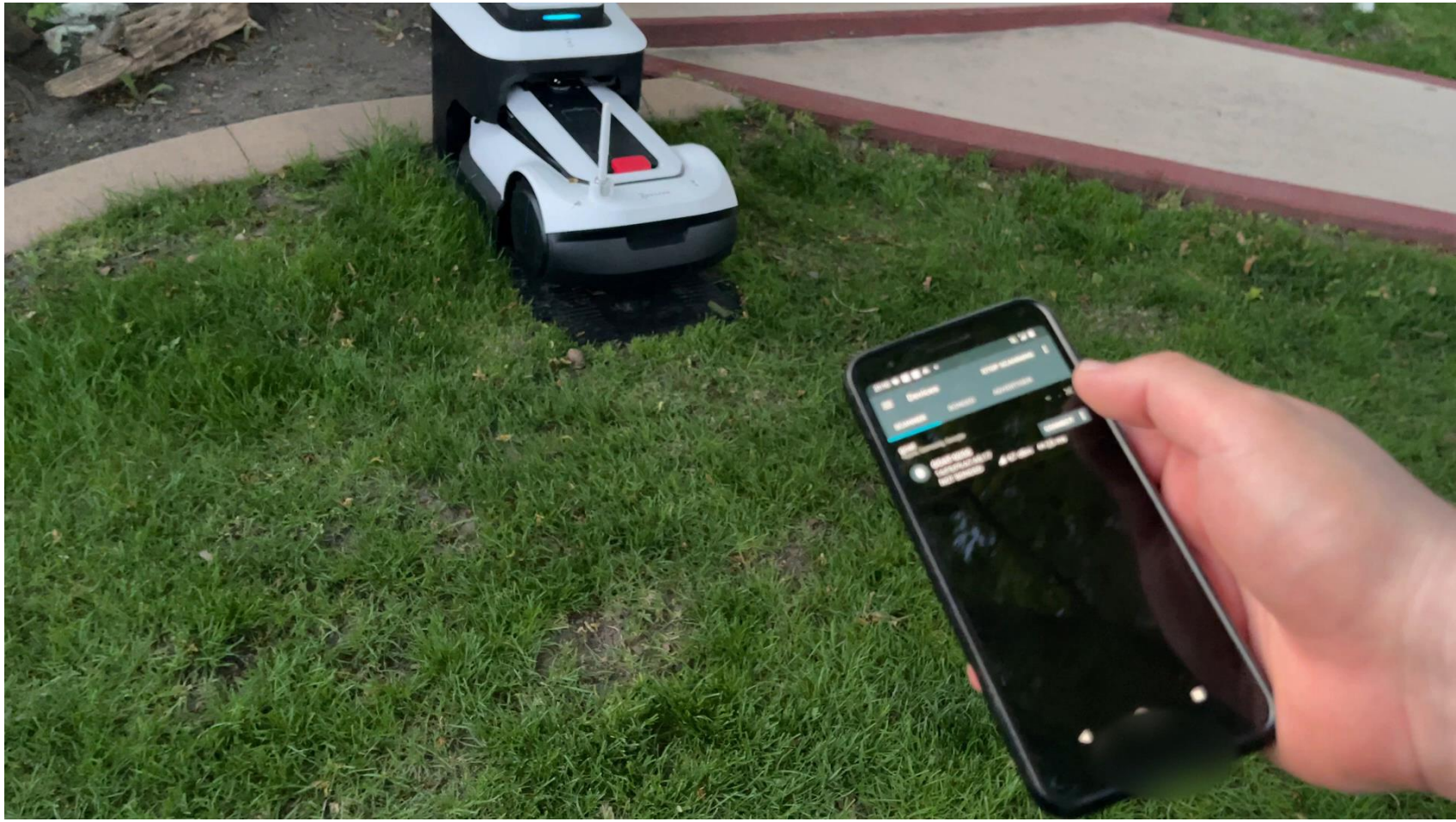
```
bar = JsonObjectGetString(v1, "foo");

snprintf(&command, 256LL, " foo =\"%s\" %s", bar , " /usr/bin/somestuff ");

(unsigned int)popen((__int64)&command, (__int64)&v16, 1024LL)
```

BLE RCE video demo (Target: Goat G1)



https://youtu.be/tV0Mh0mkXUE
https://youtu.be/YlUnoCOjE1M

BLE RCE video demo (Target: Goat G1)



https://youtu.be/tV0Mh0mkXUE
https://youtu.be/YlUnoCOjE1M

BLE RCE live demo (Target: X2)

Robot worm scenario

ROOTING

Countermeasures

- Many interesting binaries are obfuscated
  - Lots of XOR and byte shifting to hide strings
- Anti-debugging features
  - Detection of LD_PRELOAD
  - Detection of ptraces and debuggers
- SecureBoot / Android Verified Boot (AVB)
  - Enabled on some devices
  - Usage of dm-verity to protect rootfs

Root shell

- Login shell is accessible via UART
- Problem: every device has a different root password
  - Not hard-coded, set at boot time
- Function hidden and obfuscated
- Responsible program: “eco_passwd”
- Computation:

  base64(sha256({model}d4:3d:7e:fa:12:5d:C8:02:8F:0A:E2:F5{sn}\n))
- Tool: https://builder.dontvacuum.me/ecopassword.php

Firmware updates

- Firmware is encrypted but not signed
- Encryption key dynamically computed
  - Algorithm well hidden and protected against debugging
- Reverse engineering supported by „tihmstar“
- AES 128 CBC IV and Key derived from:

Format string: "vX2Z3X3RhcmdldCA%s1jdSAt%sbyBtYW4%dy5iaW4%s%x825xx%s„

% ("ECO-PT", model, section_type, "", section_len, "jeff-hk@126.com")

Persistence: RootFS modification

- Only a few models check for integrity of RootFS
- Example: Deebot X1 & Z1
  - Partitions are SquashFS packed in JFSS2
  - No signatures or verifications
  - Can be unpacked and repacked

Persistence: Autostart

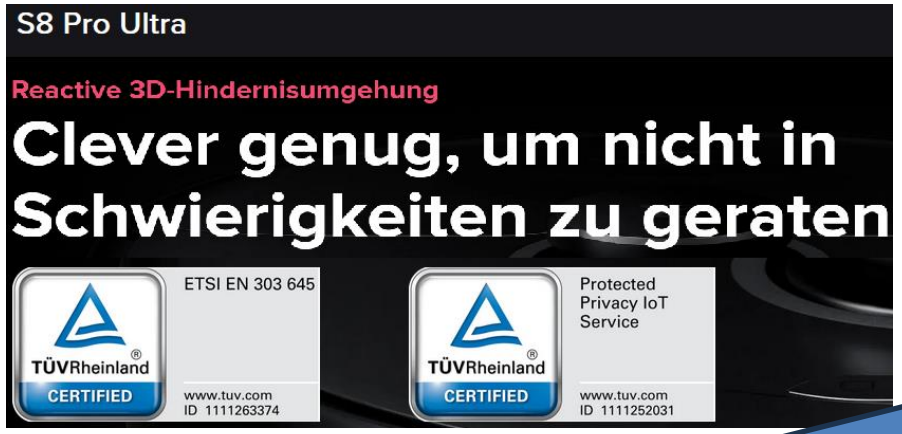
- Forgotten debugging feature
- At bootup:
  - Check if „/data/autostart“ exists
  - Run any .sh scripts in that folder
- Reminder: „/data“ not encrypted or protected in any way
- Limitation: disabled on some devices

Persistence: Factory resets

- Factory resets delete* all files from /data
- Filesystem is not recreated
- Idea: make file immutable
  - Use “chattr” to add immutable attribute
  - Immutable files will survive factory resets and updates

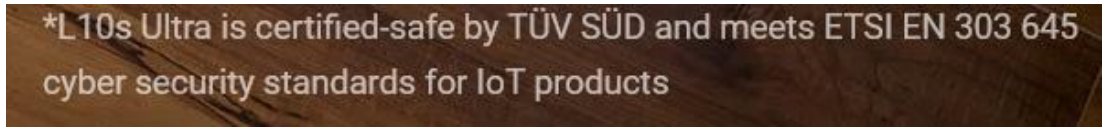
TAKE-AWAY LESSONS

Can you rely on Certifications?



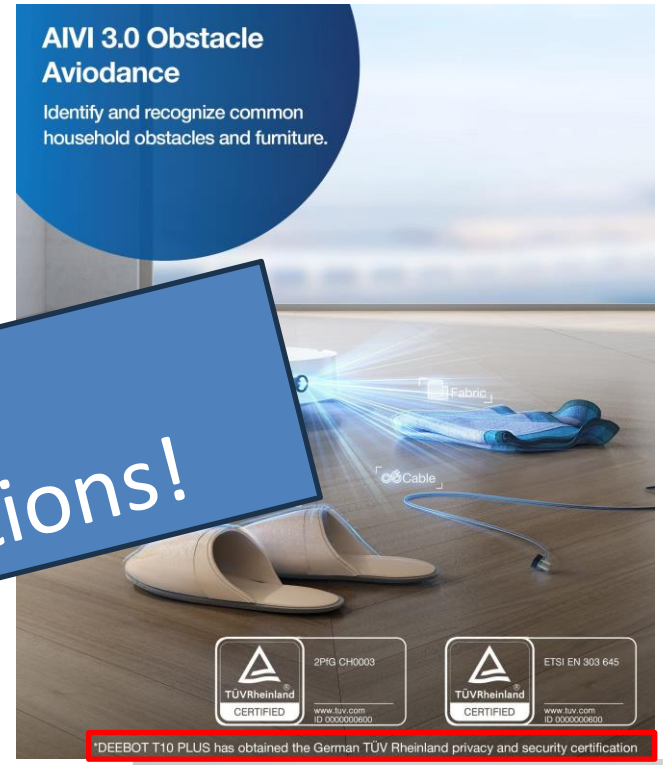
Source: https://de.roborock.com/pages/roborock-s8-pro-ultra

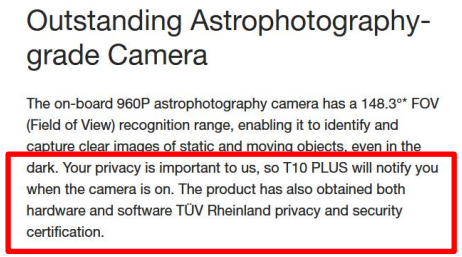
Source: https://www.mi.com/global/product/xiaomi-robot-vacuum-x10-plus/



Source: https://www.dreametech.com/products/dreamebot-l10s-ultra

Source: Ecovacs iOS application loading screen





Source: https://www.ecovacs.com/global/deebot-robotic-vacuum-cleaner/deebot-t10-plus

All Devices and Apps have been compromised regardless of certifications!

Vulnerable devices and apps

- Devices with vulnerable BLE devices
  - Impacts Goat lawnmower, X2 derived devices
  - Turn off devices when not in use
  - Wait for firmware update and fix
- Devices that do not use BLE
  - Do not update firmware if you want to keep root access
- Vulnerable apps
  - Do not connect to insecure/untrusted WiFi APs

Used devices

- Be careful with used devices
  - May come with compromised firmware
  - Difficult to verify
- Do a factory reset before selling/disposing
  - Devices contain a lot of sensitive data
  - Check the manual for a factory-reset
  - Warning: even a factory-reset leaves data behind

Choose your partners/roommate wisely

- Devices can be weaponized for stalking
- Remove shared access to accounts
- Change passwords
- When in doubt: do a factory reset and reprovision devices*

Summary

- We have root for most released Ecovacs robots
  - Usage of their UART interface and authentication
  - BLE RCE to get initial access
  - Persistence and operation of custom firmware for some
- We can validate and verify vendors claims
- There are a lot of security and privacy issues
  - Applies to App, Robots and Cloud
  - Certifications did not help to prevent them
- Work allows further research into IoT and AI

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Sören Beye (https://twitter.com/hypfer)
Tihmstar (https://twitter.com/tihmstar)
Aapo Oksman (https://twitter.com/aapooksman)

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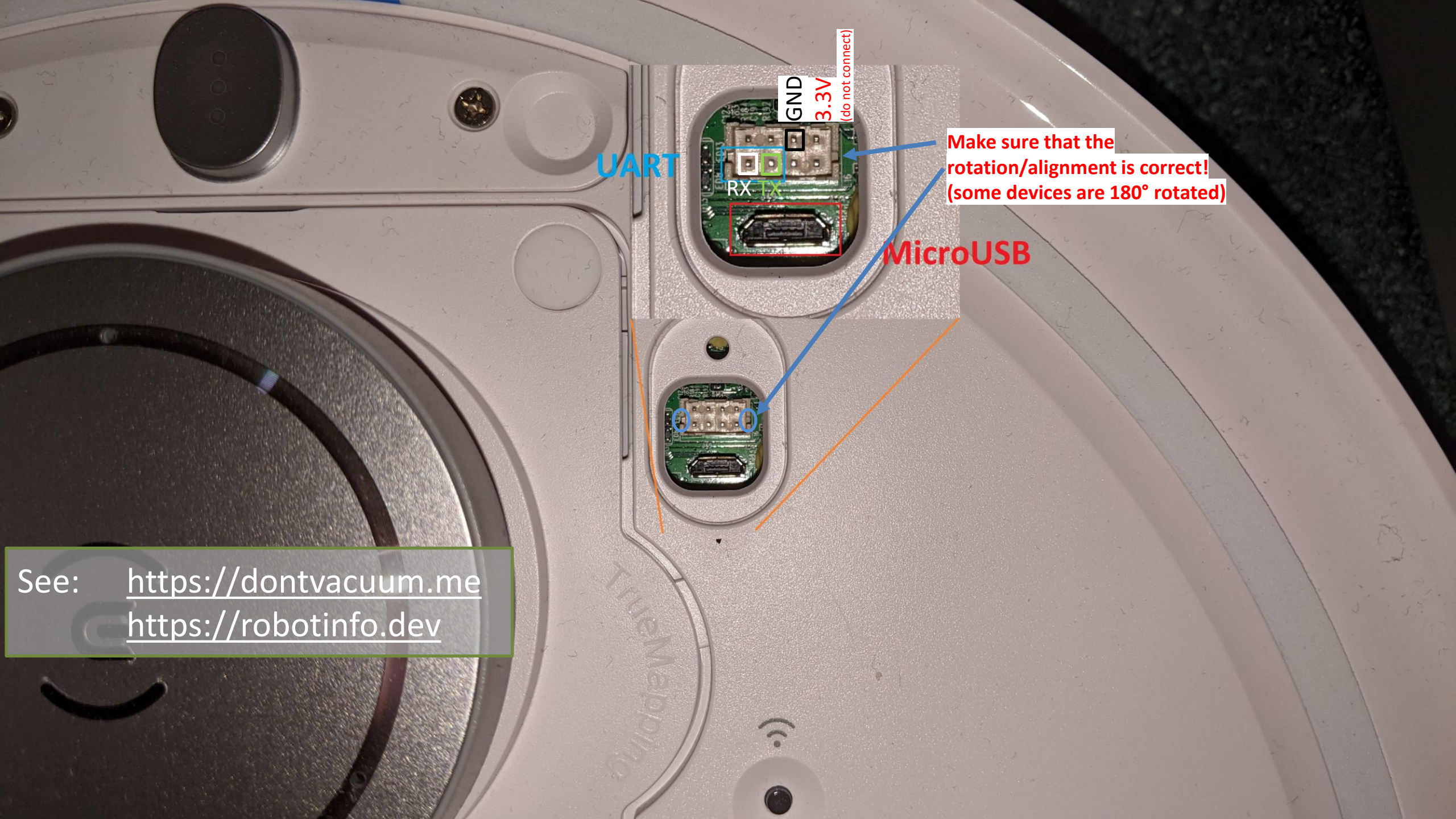
GND
3.3V
(do not connect)
UART
RX TX
Make sure that the
rotation/alignment is correct!
(some devices are 180° rotated)
MicroUSB
See: https://dontvacuum.me
https://robotinfo.dev